HASHIR AHMED

22I-1988

DS

DW-PROJECT

Project Overview:

```
Database connection established successfully!
Loading master data into memory...
Loaded 5891 customers and 3631 products
Processing transactional_data.csv using Batch HYBRIDJOIN...
HYBRIDJOIN: Joining transactional stream with customer and product master data...
Total records to process: 550068
Batch 1: HYBRIDJOIN processed 1000 records, Failed 0 records (Total: 1000/550068)
Batch 2: HYBRIDJOIN processed 1000 records, Failed 0 records (Total: 2000/550068)
Batch 3: HYBRIDJOIN processed 1000 records, Failed 0 records (Total: 3000/550068)
Batch 4: HYBRIDJOIN processed 1000 records, Failed 0 records (Total: 4000/550068)
Batch 5: HYBRIDJOIN processed 1000 records, Failed 0 records (Total: 5000/550068)
Batch 6: HYBRIDJOIN processed 1000 records, Failed 0 records (Total: 6000/550068)
Batch 7: HYBRIDJOIN processed 1000 records, Failed 0 records (Total: 7000/550068)
Batch 8: HYBRIDJOIN processed 1000 records, Failed 0 records (Total: 8000/550068)
Batch 9: HYBRIDJOIN processed 1000 records, Failed 0 records (Total: 9000/550068)
Batch 10: HYBRIDJOIN processed 1000 records, Failed 0 records (Total: 10000/550068)
Batch 11: HYBRIDJOIN processed 1000 records, Failed 0 records (Total: 11000/550068)
Batch 12: HYBRIDJOIN processed 1000 records, Failed 0 records (Total: 12000/550068)
Batch 13: HYBRIDJOIN processed 1000 records, Failed 0 records (Total: 13000/550068)
Batch 14: HYBRIDJOIN processed 1000 records, Failed 0 records (Total: 14000/550068)
Batch 15: HYBRIDJOIN processed 1000 records, Failed 0 records (Total: 15000/550068)
Batch 16: HYBRIDJOIN processed 1000 records, Failed 0 records (Total: 16000/550068)
Batch 17: HYBRIDJOIN processed 1000 records, Failed 0 records (Total: 17000/550068)
Batch 18: HYBRIDJOIN processed 1000 records, Failed 0 records (Total: 18000/550068)
Batch 19: HYBRIDJOIN processed 1000 records, Failed 0 records (Total: 19000/550068)
...
Failed records: 0
Database connection closed.
HYBRIDJOIN ETL process completed successfully!
Total records joined with master data and loaded into DW: 550068
```

This project implements a near-real-time Data Warehouse (DW) prototype for Walmart, designed to analyze shopping behavior dynamically for enabling personalized offers and promotions. The system processes transactional data streams enriched with customer and product master data using the HYBRIDJOIN algorithm.

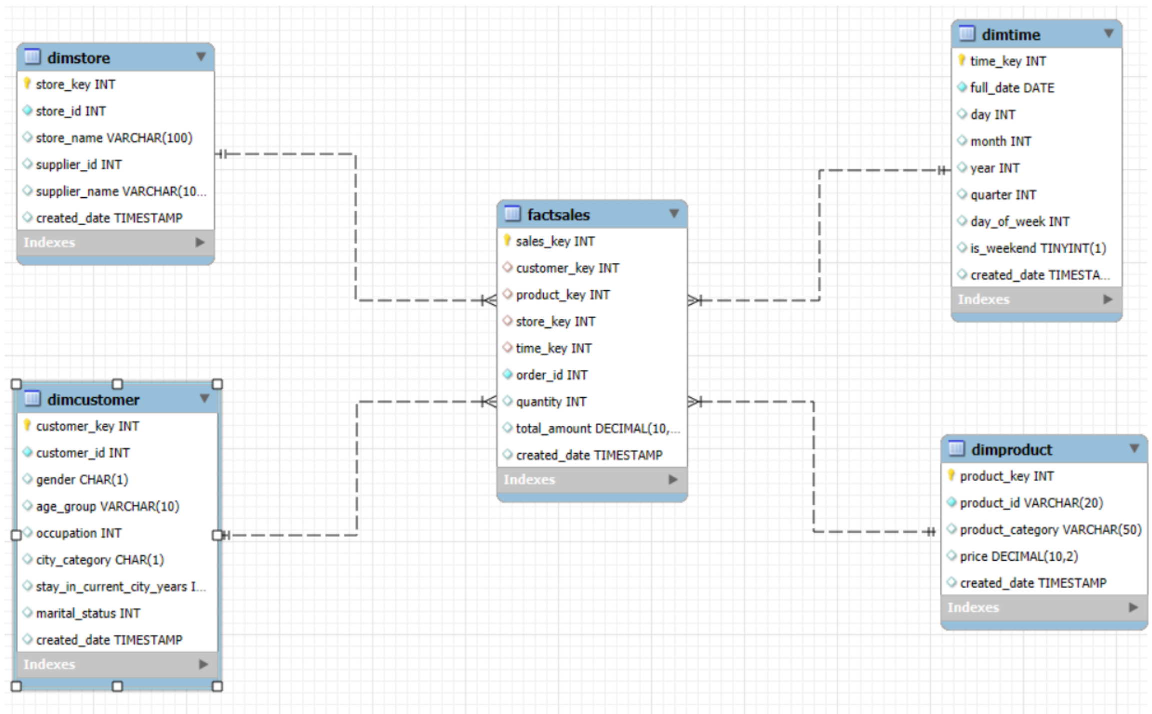
Key Objectives:

- Design and implement a star schema data warehouse
- Develop HYBRIDJOIN algorithm for stream-relation joins
- Enable near-real-time data processing capabilities
- Provide comprehensive OLAP analysis for business intelligence

Business Value:

- Real-time customer behavior analysis
- Dynamic promotion optimization
- Personalized marketing strategies
- Multi-dimensional sales analysis

Data Warehouse Schema Design:



Star Schema Components:

Dimension Tables:

- DimCustomer: Customer demographic information
- DimProduct: Product catalog and pricing
- DimStore: Store and supplier details
- DimTime: Time intelligence for trend analysis

Fact Table:

- FactSales: Sales transactions with foreign keys to all dimensions

Schema Relationships:

FactSales connects to DimCustomer, DimProduct, DimStore, and DimTime using surrogate keys.

Key Design Decisions:

- Surrogate keys used for dimensions

- Proper indexing for performance

- Complete descriptive business attributes maintained

- Referential integrity enforced through foreign key constraints

HYBRIDJOIN Algorithm Implementation:

Algorithm Components:

- Stream Buffer: Manages incoming transactional data and prevents data loss

- Hash Table: 10,000 slots for efficient tuple lookup

- Disk Buffer: Loads partitioned master data (500 tuples each)

- Queue: Doubly-linked list for FIFO and efficient deletions

Implementation Approach:

A batch-based HYBRIDJOIN was implemented using Python and data structures such as hash tables and buffers to simulate real-time data enrichment.

ETL Process:

- Extract: Read transactional data from CSV

- Transform: Enrich using HYBRIDJOIN

- Load: Insert enriched records into the DW

Shortcomings of HYBRIDJOIN

1 Memory Intensive Operations

Master data loaded entirely into RAM can be problematic for large datasets and limited-memory systems.

2 Batch Processing Limitations

Batch processing introduces latency and may not support high-velocity streams.

3 Complexity in Error Handling

Complex data structures make debugging and thread safety difficult.

Lessons Learned

1 Data Warehouse Design Principles

- Star schema improves OLAP performance

- Indexing is essential

- Dimensions must include complete descriptive attributes

2 Stream Processing Challenges

- Real-time needs efficient resource handling

- Data enrichment increases complexity

- Batch processing can simulate streaming effectively

3 Algorithm Insights

- HYBRIDJOIN balances memory-disk usage

- Data structure selection greatly affects performance

- Testing at realistic scale is critical

4 Business Intelligence Value

OLAP queries generated insights for customer segmentation, seasonal trends, product performance, and store comparisons.

Implementation Challenges & Solutions

Challenge 1: Data Quality Issues

Solution: Added validation and error checking.

Challenge 2: Performance Optimization

Solution: Applied indexing and optimized joins.

Challenge 3: Algorithm Complexity

Solution: Simplified HYBRIDJOIN using batch mode.

Business Insights Generated

- Identify high-value customer demographics

- Analyze product performance trends

- Detect seasonal purchasing behavior

- Compare store performance

- Identify cross-selling opportunities

Queries:

1

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product_id	product_category	month	year	day_type	total_revenue	revenue_rank
P00052842	Automotive	1	2020	weekday	5825.90	1
P00220442	Health & Beauty	1	2020	weekday	5409.00	2
P00031042	Toys	1	2020	weekday	5353.48	3
P00059442	Household Essentials	1	2020	weekday	4966.14	4
P00044442	Grocery	1	2020	weekday	4883.34	5

2

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

gender	age_group	city_category	total_purchase_amount	total_orders	avg_order_value
M	26-35	B	13309614.86	70147	81.228997
M	26-35	A	10660062.32	56254	81.204674
M	26-35	C	7986609.01	42434	80.622328
M	36-45	B	6814356.17	36488	80.195313
M	18-25	B	6051488.27	31561	81.803399

3

Result Grid | Filter Rows: | Export: | Wr

product_category	occupation	total_sales	total_orders
Appliances	0	32746.96	442
Appliances	1	32688.07	391
Appliances	7	29753.63	376
Appliances	4	29738.37	418
Appliances	20	23230.61	287

4

gender	age_group	quarter	year	quarterly_purchases	prev_quarter_purchases
F	0-17	1	2020	42497.75	NULL
F	0-17	2	2020	39753.65	42497.75
F	0-17	3	2020	42652.15	39753.65
F	0-17	4	2020	43640.88	42652.15
F	18-25	1	2020	202085.32	NULL

5

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product_category	occupation	total_sales	occupation_rank
Appliances	0	32746.96	1
Appliances	1	32688.07	2
Appliances	7	29753.63	3
Appliances	4	29738.37	4
Appliances	20	23230.61	5

6

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

city_category	marital_status	month	year	monthly_sales	order_count
A	1	6	2020	4477.77	61
A	0	6	2020	5703.66	75
B	1	6	2020	9993.29	117
B	0	6	2020	11128.65	142
C	0	6	2020	7538.07	89

7

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

stay_in_current_city_years	gender	avg_purchase_amount	total_orders
0	F	81.633192	39711
0	M	80.995326	134094
1	F	80.588889	119752
1	M	80.828701	332714
2	F	81.600887	56959

8

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

city_category	product_category	total_revenue	city_rank
B	Appliances	113350.37	1
A	Appliances	87216.40	2
C	Appliances	75566.59	3
B	Arts, Crafts & Sewing	721749.48	1
A	Arts, Crafts & Sewing	516994.78	2

9

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product_category	year	month	monthly_sales	prev_month_sales	growth_percentage
Appliances	2020	1	3253.67	NULL	NULL
Appliances	2020	2	2151.23	3253.67	-33.88
Appliances	2020	3	2900.06	2151.23	34.81
Appliances	2020	4	5112.07	2900.06	76.27
Appliances	2020	5	5922.03	5112.07	15.84

10

Result Grid | Filter Rows: | Export:

age_group	day_type	total_sales	total_orders
0-17	weekday	344021.21	4185
0-17	weekend	145230.58	1808
18-25	weekday	2273674.57	27700
18-25	weekend	897749.84	10992
26-35	weekday	4937362.64	60681

11

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product_id	product_category	month	year	day_type	total_revenue	revenue_rank
P00052842	Automotive	1	2020	weekday	5825.90	1
P00220442	Health & Beauty	1	2020	weekday	5409.00	2
P00031042	Toys	1	2020	weekday	5353.48	3
P00059442	Household Essentials	1	2020	weekday	4966.14	4
P00044442	Grocery	1	2020	weekday	4883.34	5

12

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

store_name	year	quarter	quarterly_revenue	prev_quarter_revenue	growth_rate
Electro Mart	2017	1	622491.01	NULL	NULL
Electro Mart	2017	2	594539.07	622491.01	-4.49
Electro Mart	2017	3	585175.34	594539.07	-1.57
Electro Mart	2017	4	619155.13	585175.34	5.81
Game Zone	2017	1	858741.09	NULL	NULL

13

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

store_name	supplier_name	product_name	product_category	total_sales	order_count
Electro Mart	Sony Corporation	P00220442	Health & Beauty	431133.36	3042
Electro Mart	Sony Corporation	P00085942	Electronics	269932.00	2277
Electro Mart	Sony Corporation	P00271142	Health & Beauty	226545.12	1857
Electro Mart	Sony Corporation	P00255842	Arts, Crafts & Sewing	206530.54	3266
Electro Mart	Sony Corporation	P00286642	Patio & Garden	201121.60	1281

14

Result Grid | Filter Rows: | Export: | Wrap Cell Cont

product_id	product_category	season	seasonal_sales	order_count
P00000142	Home & Kitchen	fall	38669.80	675
P00000142	Home & Kitchen	spring	35611.80	626
P00000142	Home & Kitchen	summer	40393.40	736
P00000142	Home & Kitchen	winter	35889.80	660
P00000242	Electronics	fall	16824.64	200

15

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

store_name	supplier_name	year	month	monthly_revenue	prev_month_revenue	volatility_percentage
Pakistan	Pakistan	2018	4	22102.63	14805.00	49.29
Pakistan	Pakistan	2019	8	22175.85	16200.42	36.88
Pakistan	Pakistan	2018	10	19190.34	14154.51	35.58
Pakistan	Pakistan	2020	5	19676.19	14925.52	31.83
Pakistan	Pakistan	2017	2	14721.31	21573.55	-31.76

17

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

store_name	supplier_name	product_category	year	total_revenue
Electro Mart	all suppliers	all products	NULL	14460385.88
Electro Mart	Sony Corporation	all products	NULL	14460385.88
Electro Mart	Sony Corporation	Appliances	NULL	80349.84
Electro Mart	Sony Corporation	Appliances	2015	14111.01
Electro Mart	Sony Corporation	Appliances	2016	14338.19

18

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

product_id	product_category	half_year	half_year_revenue	half_year_quantity	half_year_orders
P00000142	Home & Kitchen	h1	11286.80	406	215
P00000142	Home & Kitchen	h2	16263.00	585	279
P00000142	Home & Kitchen	year	27549.80	991	494
P00000242	Electronics	h1	6609.68	154	78
P00000242	Electronics	h2	7468.08	174	85

19

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

product_id	product_category	full_date	daily_sales	avg_daily_sales	sales_status	percentage_above_avg	anomaly_explanation
P00110842	Grocery	2018-03-23	2227.66	444.291777	spike	401.40	major promotional event or data error
P00059442	Household Essentials	2016-03-06	2147.52	422.000058	spike	408.89	major promotional event or data error
P00351342	Grocery	2018-07-12	2055.24	400.981014	spike	412.55	major promotional event or data error
P00112542	Grocery	2016-12-26	2010.69	360.491038	spike	457.76	holiday season peak
P00192542	Grocery	2020-10-19	1894.20	308.075594	spike	514.85	major promotional event or data error

Conclusion

This project showcases a near-real-time data warehouse implementation using HYBRIDJOIN. It enriches transactional data streams, loads them into a star-schema DW, and enables multi-dimensional OLAP analysis. Walmart can use this system for personalized marketing, dynamic promotions, and enhanced business decision-making.